

Meridian Sorbent Technologies

Process Economics Group

Adsorbent costing and stage-gate policy

Policy ref. MST-PE-2018-06 | issued 5 October 2018 | applies to all sorbent candidates at stage gate

1. Benchmark table costing

A candidate is entered in the adsorbent benchmark table at its cash production cost per kilogram of released product. Cash cost is every line of the pilot record other than equipment depreciation, which is a plant-level allocation and is not counted. Only batches dispositioned Released enter this figure and the capacity quoted beside it. The figures are not averaged across the whole campaign.

2. Manufacturing-loss sensitivity

Every candidate is also shown on a manufacturing-loss basis: the cash cost (as in section 1) of every batch in the campaign, released or rejected, carried by the released product alone. This is a stage-gate sensitivity. It does not replace the benchmark table entry.

3. Gate tests

- (a) Efficiency. On both the benchmark-table basis and the manufacturing-loss basis, the candidate must match or exceed the efficiency of its closest benchmark.
- (b) Thermal margin. The handling or regeneration ceiling of the recovered material, determined as in the laboratory's thermal screening note, must be at least 25 °C above the regeneration temperature assumed in the benchmark costing.
- (c) Evidence. Guidance from trade groups, suppliers or other external sources does not count towards a test unless it was derived from the candidate's own material. Where such guidance and the laboratory's own characterisation of the candidate differ, the laboratory's data governs. Performance claimed over repeated regeneration cycles counts only where cycles have been demonstrated on the candidate.

4. Outcome

A candidate that passes every test proceeds to the next stage. A candidate that fails any test, or whose test cannot be completed from the evidence available, is held for validation until the failed or incomplete test is resolved.